

Advanced Econometrics

Introduction to Time Series

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Stationarity (What and Why)

What is stationarity?

A time series is **stationary** if its statistical properties stay **constant over time**. This means the data behaves in a stable and predictable way.

Formally:

- Constant **mean** (no upward/downward trend)
- Constant **variance** (no changing volatility)
- Constant **autocorrelation** structure over time

Why is it important?

- Many models (e.g., **ARIMA**) assume stationarity
- Makes relationships in data **stable and learnable**
- Prevents **misleading patterns** caused by trends or seasonality
- Improves **forecast accuracy**

If data is not stationary:

- Trends or seasonality dominate the signal
- Model predictions become unreliable

How to fix it:

- Differencing (subtract previous values)
- Detrending / removing seasonality

Stationarity Examples (Very Simple)

a) White noise is stationary

- Random values (no pattern)
- Mean = constant (usually 0)
- Variance = constant
- No trend, no structure
- $\Rightarrow$ **Stationary**

b) Random walk is non-stationary

- Each value = previous value + random noise
- Values **drift over time**
- Variance increases
- No stable mean
- $\Rightarrow$ **Non-stationary**

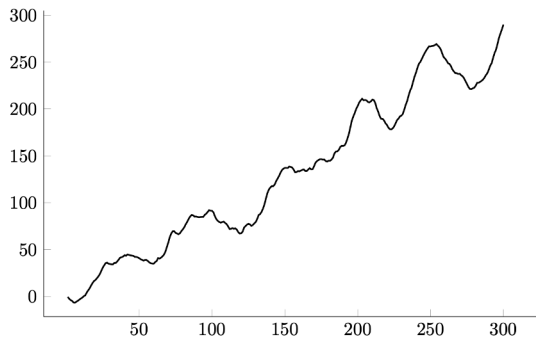
c) How to assess stationarity (for exercises)

- Is there a **trend**? $\Rightarrow$ not stationary
- Is variance changing? $\Rightarrow$ not stationary
- Does it look stable over time? $\Rightarrow$ stationary

Shortcut: Stable = good (stationary), drifting = bad (non-stationary)

Example 1

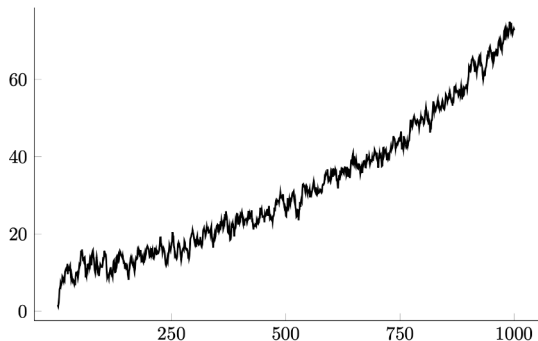
Example 1



- Strong, persistent **upward trend**
- Mean rises from 0 to ≈ 300
- Classic **random walk** behaviour
- $y_t \sim I(1)$ — difference to fix

Example 2

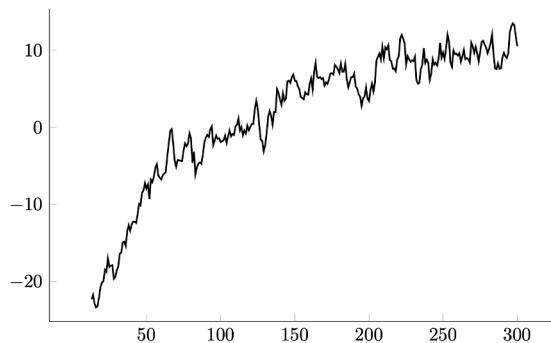
Example 2



- Smooth, steady **upward trend**
- Mean grows from 0 to > 70
- Variance also **grows with time**
- $y_t \sim I(1)$

Example 3

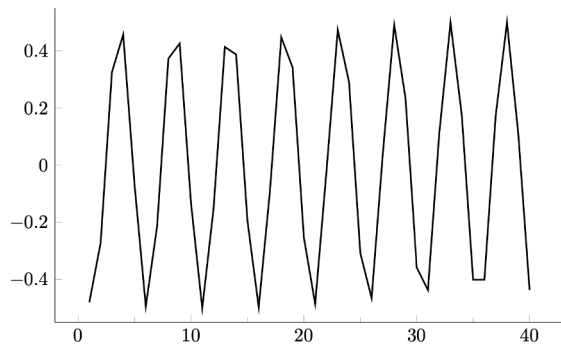
Example 3



- Starts at -25 , ends near $+12$
- Clear **upward drift** throughout
- Mean is **not constant**
- $y_t \sim I(1)$

Example 4

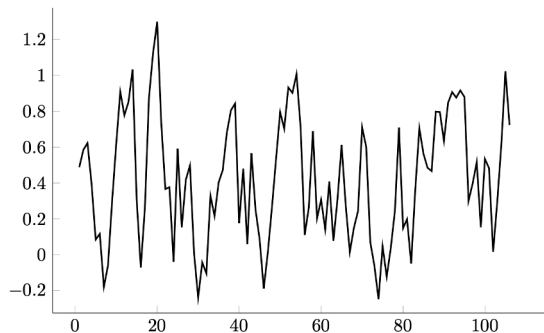
Example 4



- **Regular oscillation** around zero
- Constant amplitude ($\approx \pm 0.5$)
- No trend, constant variance
- $y_t \sim I(0)$ 51

Example 5

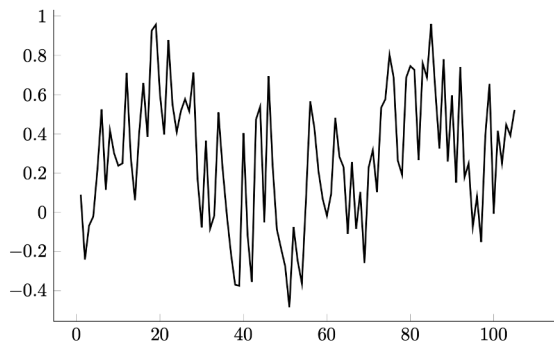
Example 5



- Fluctuates around **constant mean** (≈ 0.4)
- No visible trend
- Variance stable throughout
- $y_t \sim I(0)$ 51

Example 6

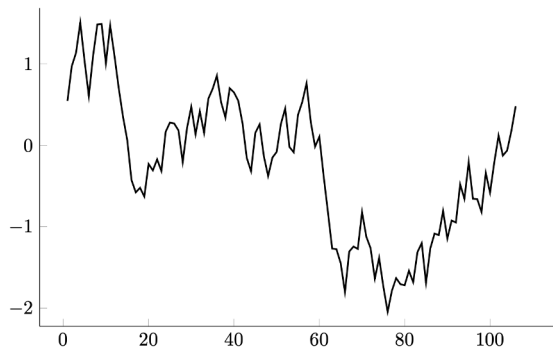
Example 6



- Bounces around **constant mean** (≈ 0.3)
- No systematic drift
- Stable variance throughout
- $y_t \sim I(0)$ 51

Example 7

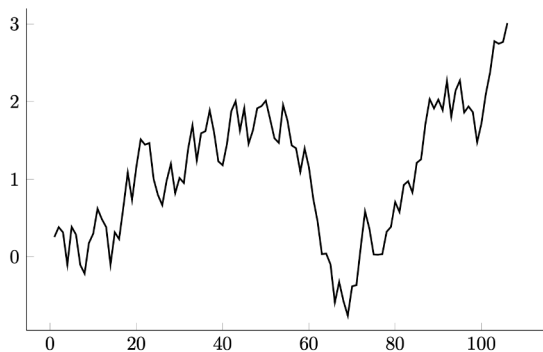
Example 7



- Starts high, **dips to** -2 , recovers
- Mean is clearly **not constant**
- Series wanders — random walk behaviour
- $y_t \sim I(1)$

Example 8

Example 8



- **Wave / hump pattern** — rises then falls
- Mean shifts systematically over time
- Not fluctuating around fixed level
- $y_t \sim I(1)$

Summary: Examples 1–8 (probably)

Ex.	Verdict	Key reason
1	NOT stationary	Strong upward trend, $0 \rightarrow 300$
2	NOT stationary	Smooth upward trend, growing variance
3	NOT stationary	Drifts from -25 to $+12$
4	Stationary	Regular oscillation, mean = 0
5	Stationary	Bounces around ≈ 0.4
6	Stationary	Bounces around ≈ 0.3
7	NOT stationary	Wanders (dip to -2 , then recovery)
8	NOT stationary	Wave/hump pattern
NOT stationary: 5 cases		Stationary: 3 cases

Drift vs Trend

Both mean “going up or down”... but not the same!

Deterministic Trend (TREND)

- Smooth, predictable increase/decrease
- Same pattern every time
- Example:

$$y_t = \beta t + \varepsilon_t$$

- $\Rightarrow$ Remove it by subtracting the trend

Drift (in Random Walk)

- Small constant push each step + randomness
- Still **random and wandering**

$$y_t = y_{t-1} + c + \varepsilon_t$$

- $\Rightarrow$ Path is unpredictable (not a smooth line!)

Key difference

- Trend = **predictable line**
- Drift = **random walk with upward bias**

Memory trick:

- Trend = “straight line”
- Drift = “drunk walk going uphill”

Deterministic vs Stochastic Trend

Deterministic Trend (predictable)

- Smooth, fixed pattern (like a straight line)
- Same shape every time

$$y_t = \beta t + \varepsilon_t$$

- Shock (ε_t) = **temporary**
- Always goes back to the trend
- $\Rightarrow$ Easy to remove (just subtract trend)

Stochastic Trend (random)

- Random walk $\rightarrow$ unpredictable path
- Example:

$$y_t = y_{t-1} + \varepsilon_t$$

- Shock = **permanent effect**
- No “return” to a line
- $\Rightarrow$ Must difference the data

Key difference

- Deterministic = **predictable + temporary shocks**
- Stochastic = **random + permanent shocks**
- Deterministic = “train on tracks”
- Stochastic = “drunk person walking”

Stationarity Tests

Big question: why do we test?

- Data can look non-stationary because:
 - It has a **trend** (deterministic)
 - It has a **unit root** (random wandering)
- We need to know which one!

Unit root (very simple idea):

- Data behaves like a **random walk**
- Shocks have **permanent effects**
- $\Rightarrow$ **Non-stationary**

ADF Test (Augmented Dickey-Fuller):

- Main test for stationarity
- Checks: **Is there a unit root?**

Decision rule (IMPORTANT):

- Null hypothesis: **Unit root exists** (bad $\rightarrow$ non-stationary)
- If $p\text{-value} < 0.05 \Rightarrow$ reject $\rightarrow$ **stationary**
- If $p\text{-value} > 0.05 \Rightarrow$ fail to reject $\rightarrow$ **non-stationary**

Why this matters (Nelson & Plosser, 1982):

- Many economic series behave like **random walks**
- Shocks are **permanent**, not temporary
- $\Rightarrow$ must difference the data before modeling

Why Differencing Helps

Problem: non-stationary data

- Data has a **trend** or **random walk**
- Mean is changing over time
- Models get confused

Idea of differencing

- Instead of using y_t , use:

$$\Delta y_t = y_t - y_{t-1}$$

Why this works

- Removes **trend** (turns slope into flat)
- For random walk:
 - $y_t = y_{t-1} + \varepsilon_t$
 - $\Rightarrow \Delta y_t = \varepsilon_t$ (just noise!)
- Mean becomes **constant**
- Variance becomes **stable**

Intuition:

- Original: **position** (keeps drifting)
- Differenced: **change** (stable)

Shortcut: Differencing = removes drift $\rightarrow$ makes data stationary

Dickey-Fuller Test I, II, III

Goal of Dickey-Fuller: Check if a time series is:

- **stationary**
- or has a **unit root** (random walk, non-stationary)

Type I: no constant, no trend

- Data fluctuates around **0**
- **Real-life example:**
 - Daily measurement errors
 - Noise in sensors (around zero)

Type II: constant, no trend

- Data fluctuates around a **fixed level**
- **Real-life example:**
 - Temperature in a controlled room
 - Heart rate at rest

Type III: constant + trend

- Data follows a **trend over time**
- **Real-life example:**
 - GDP growth over years
 - Population increase

Thank you